For Examiner's Use

6 **(a)** Rutherford's α-particle scattering experiment was conducted to probe the structure of the atom.

Explain why only a small proportion of the α-particles incident on the gold foil are deflected through angles greater than 90°.

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…… [3]

(b) A deuterium nuclear fusion reaction is given below:

$$^{2}_{1}H + ^{2}_{1}H \rightarrow A + B$$

where *A* is a charged particle and *B* is an unknown element.

The process is followed by a reaction involving *B* with another deuterium atom given below:

$$^{2}_{1}H + B \rightarrow ^{4}_{2}He + ^{1}_{0}n$$

<u>Atomic masses</u>
Deuterium $^{2}_{1}H$ atom : 2.01410 *u*
Protium $^{1}_{1}H$ atom : 1.00783 *u*
Neutron $^{1}_{0}n$: 1.00867 *u*

(i) Deduce the nucleon and proton numbers of A and B.

nucleon number of A = ……

proton number of A = ……

nucleon number of B = ……

proton number of B = ……
[2]

[Turn over

For Examiner's Use

(ii) Explain why the above nuclear fusion reactions cannot occur with slow moving particles of deuterium nuclides.

……

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…… [1]

(iii) Determine the binding energy per nucleon of a $^{2}_{1}H$ nuclide.

binding energy per nucleon = …… MeV [4]

(c) Sodium-24, a radioactive isotope with a *half-life* of 15 hours, is used for the study of electrolytes within the human body.

A small volume of a solution containing Sodium-24 had an activity of 12000 disintegrations per minute when it was injected into the bloodstream of a patient. After 39 hours, the activity of 1.0 cm^3 of the blood was found to have 0.50 disintegrations per minute.

(i) Explain what is meant by the *half-life* of Sodium-24.

……

……

…… [1]

For Examiner's Use

(ii) Determine the volume of blood in the patient.

volume = …… cm^3 [3]

(iii) Suggest why Sodium-24 is suitable for use in human patients.

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